

Smart Waste Management System for Sustainable Cities

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Abstract - Rapid urbanization and changing consumption patterns are increasing the volume and complexity of municipal solid waste, straining conventional collection and disposal practices, and contributing to bin overflow, illegal dumping, and low recycling rates. This paper proposes an integrated smart waste management framework that combines multimodal AI for disposal guidance, a Pay-As-You-Throw-inspired fee mechanism informed by LSTM based waste prediction and XGBoost based tax regression, a bilingual (English/Sinhala) RAG based conversational assistant over municipal documents, and a temporal LSTM based forecasting framework for ward-level waste demand prediction, integrated with resource allocation and route optimization for efficient municipal waste collection. Experiments on real world inspired datasets show strong waste-classification performance, accurate weekly waste prediction, improved response relevance in the conversational agent, and superior forecasting accuracy relative to a purely temporal LSTM baseline, indicating the potential of the proposed system to support sustainable, data-driven urban waste management.

Keywords- *Smart Waste Management, Multimodal AI, Behavior Based Taxation, Conversational Agents, Municipal Solid Waste, LSTM, XGBoost.*

I. INTRODUCTION

Municipal solid waste management has become a critical challenge in rapidly growing urban cities, where increasing waste generation, poor segregation practices, bin overflow, illegal dumping, and inefficient fixed-route collection directly affect public health, environmental quality, and municipal service efficiency. Conventional waste management systems often depend on static collection

schedules, limited real time bin monitoring, and manual decision making, which can lead to unnecessary collection trips, delayed responses to high-demand areas, increased fuel and labor costs, and low recycling efficiency. At the household level, citizens often lack clear guidance on proper waste disposal, while existing fee mechanisms provide limited motivation for responsible waste reduction and segregation.

To address these issues, this research proposes an integrated smart waste management system that combines AI-based disposal guidance, IoT-enabled bin monitoring, behavior based smart taxation, bilingual conversational support, and LSTM based waste demand forecasting with resource allocation and route optimization. The societal impact of this system lies in improving waste segregation, reducing overflow, encouraging sustainable citizen behavior, and supporting more efficient municipal collection planning. From a scientific perspective, the research contributes by integrating multimodal AI, machine learning, retrieval-augmented generation, and predictive analytics into a unified framework for sustainable urban waste management. This integrated approach supports data-driven decision making for both citizens and municipal authorities, making it suitable for smart and sustainable city environments.

II. RELATED WORKS

Existing works commonly represents smart household facing waste support in three streams. automated waste classification, IoT based bin monitoring with collection optimization, and time-series prediction of waste

generation and bin fill levels. In many systems, computer vision or text-based models are used to identify the waste category to support correct separation. [1]. However, these approaches often stop at category prediction and do not provide user-specific technique selection (e.g., reuse/recycle/upcycle) together with practical, step-by-step disposal guidance. Another widely used direction equips bins with sensors and connectivity to estimate fill levels and enable more efficient disposal planning. [2], but such solutions are frequently designed from the municipal operations perspective and provide limited assistance at the moment of disposal. Forecasting oriented studies use temporal models (including recurrent neural networks) to predict future waste generation or bin status to support proactive planning. [3], yet purely temporal predictors may ignore spatial interactions between areas that influence localized overflow risk. Building on these common foundations, this combines multimodal identification with user-centered guidance and operational capacity/overflow prediction to support correct disposal decisions better and reduce overflow events.

In parallel, the economic research on municipal waste management has promoted Pay-As-You-Throw (PAYT) and similar pricing systems based on volume or weight to encourage waste reduction and better source separation. [4], [5], [6]. In practice, these programs usually operate with fixed tariff schedules based on observed waste amounts or standard container sizes. [5], [6]. While these designs can lower residual waste, typical PAYT programs seldom use predictive analytics or targeted behavioral segmentation. [4], [5]. As a result, short-term changes and ongoing high-waste behavior may be charged the same, and historical data may not be fully used when setting tariffs that are fair and responsive. [4].

Conversational systems for waste management typically take the form of static FAQ portals or rule-based chatbots maintained by municipal authorities. These systems can provide basic information about collection schedules and allowed materials, but are often limited in conversational flexibility and language coverage. [7]. With the emergence of large language models (LLMs), interest has grown in deploying AI-based assistants for public services. [8]. However, in the absence of robust grounding in official documents and strict constraints, such systems may generate incorrect or non-compliant information. This risk is particularly salient in domains where local regulations, bin color codes, and fee rules vary by jurisdiction and must be communicated precisely in local languages.

Spatio-temporal modeling is essential for forecasting zone-level waste demand in municipal systems. In the EcoFit framework, temporal patterns in historical waste data are

captured using Long Short-Term Memory (LSTM) networks, which effectively model seasonal trends and short-term fluctuations in municipal waste generation. These temporal predictions are derived from multiple municipal data sources, including service logs, complaint records, operational indicators, and environmental factors, enabling more reliable demand estimation across wards. Similar temporal forecasting approaches have demonstrated strong performance in urban demand prediction tasks such as traffic and city service management. [9], [10]. Based on the predicted demand risk levels, the system generates a prioritized operational plan that supports demand scoring, resource allocation, and route optimization for municipal waste collection. By translating predictive insights into actionable service planning, the framework facilitates data-driven resource allocation and operational decision-making for municipalities. [11].

The framework proposed in this paper extends prior work by tightly integrating multimodal AI for household guidance, predictive behavior-based tariffs, document-grounded conversational assistance, and LSTM based waste demand forecasting, combined with operational planning mechanisms such as demand scoring, resource allocation, and route optimization into a unified architecture. It explicitly targets both the citizen-facing and municipal-planning aspects of waste management, with an emphasis on explainability, localization, and operational applicability.

III. METHODOLOGY

A. System Architecture Overview

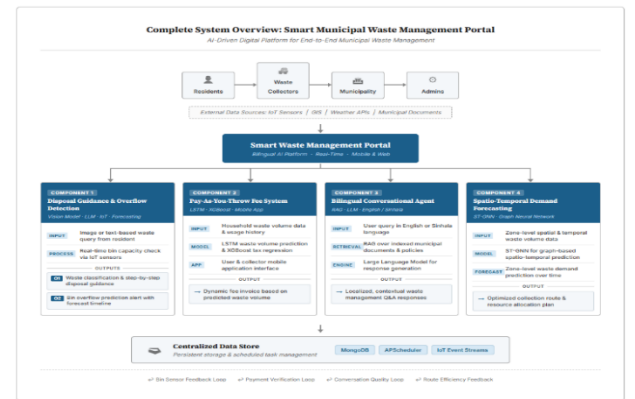


Figure 1 System Architecture Diagram

The proposed system follows a multi-layer architecture that connects data collection, AI models, and user-facing applications. As shown in Figure 1, resident inputs, IoT bin data, and municipal records are processed through four main modules: AI-based disposal guidance, LSTM

XGBoost smart fee prediction, bilingual RAG-based citizen support, and LSTM-based ward demand forecasting. These outputs are delivered through a resident mobile app, collector app, and municipal planning dashboard, enabling disposal guidance, bin-fit decisions, tax monitoring, resource allocation, and route optimization within one integrated platform.

B. Integrated System Components

1. End-to-End Disposal Guidance with Capacity Checking and Overflow Forecasting

This component delivers end-to-end disposal guidance by allowing users to submit a waste image or text description, which is classified into the correct waste category. Based on the category and user context, the system recommends the most suitable technique such as reuse, recycle, or upcycle, followed by step-by-step disposal guidance [12]. It also checks whether the selected smart bin has enough remaining capacity using IoT sensor readings converted into fill-level estimates [13], [14]. Finally, a lag-based time-series forecasting model predicts possible future overflow, helping users make correct disposal decisions and supporting municipalities in reducing bin overflow risk [15].

2. Household Waste-Based Smart Tax or Fee Calculation System

This component introduces a transparent behavior-based smart fee model by combining LSTM-based household waste forecasting with XGBoost based tax prediction. LSTM is used to forecast weekly waste quantities and handle temporal waste patterns, while XGBoost estimates tax values from predicted or recorded waste data [16], [17], [20]. Since fine grained household data may be unavailable, aggregated waste records are converted into realistic household level data using a simulation approach [18]. The final tax is calculated using a deterministic formula to keep the process fair, reproducible, and auditable, while mobile app features such as rankings, rewards, and collector verification encourage responsible waste behavior [19], [21].

3. Conversational Agent for Waste Issue Resolution and Service Guidance.

This component provides a bilingual RAG-based conversational assistant that gives citizens accurate and localized waste management information from official municipal documents [22]. The system extracts and indexes document content using FAISS for semantic retrieval and BM25 for keyword matching, especially for Sinhala location names and collection schedules. User queries are detected as English or Sinhala, then relevant document segments are retrieved, reranked, and filtered

according to the user's registered location. English responses are generated using LLaMA 3.1 8B, while Sinhala responses use a QLoRA-tuned Gemma model [23]. By grounding answers in official documents, the assistant reduces misinformation, supports questions on collection days, segregation rules, and complaints, and lowers routine inquiry workload for municipal staff [24].

4. Urban Service Demand Forecasting and Resource Optimization Using Temporal Machine Learning

This component supports city-level waste planning by using an LSTM model to predict ward level waste demand from complaint records, service logs, bin clearance data, estimated waste volume, and weather factors. The predicted demand is converted into risk categories and demand scores, helping municipal planners identify wards that need urgent attention. Based on these scores, the system allocates crews and collection resources according to priority and workload. A route optimization module then generates efficient collection routes by considering depot locations, ward priorities, and travel distance, while a web dashboard visualizes risk maps, service queues, and recommended resource deployment for better data-driven municipal planning.

IV. RESULTS

The integrated system was evaluated module-wise to assess the feasibility and effectiveness of the proposed architecture. Overall, the results indicate robust performance across system functions. In the disposal guidance pipeline, the EfficientNetB0 based image classifier showed the training accuracy behavior of the waste classification model, indicating that the model learns category-level waste features effectively during training in Figure 2.

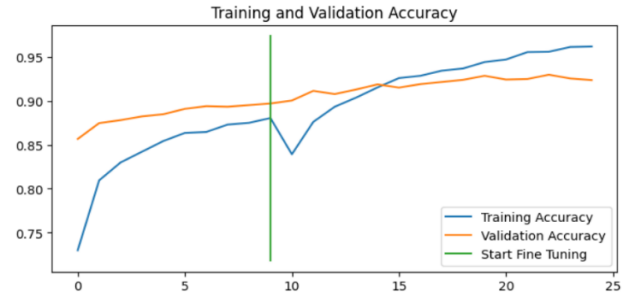


Figure 2 Model Training Accuracy Graph

The text-based classifier also performed well on descriptive queries, with most errors arising from generic or ambiguous inputs. In addition, the XGBoost-based technique recommendation model demonstrated reliable

prediction of whether reuse, recycling, or upcycling was most suitable for a given item and user profile. Finally, scenario-based evaluations that coupled classification with IoT derived bin status suggested fewer simulated overflow events than a non-adaptive baseline. The key model performances are summarized in Table 1, highlighting strong results for text classification, technique prediction, and overflow prediction within the disposal guidance component

Task/Model	Metric	Value
Text classification (API model)	Accuracy	0.98
Technique prediction (XGBoost)	Accuracy	0.85
Overflow prediction (Random Forest)	MAPE	8.37%

Table 1 Models Metrics Values

In the Smart tax calculation system, the LSTM models accurately predicted weekly household waste. They showed low RMSE and MAE along with high R^2 values. These results captured seasonal patterns and household behaviors, consistent with earlier LSTM based forecasting studies [16], [17]. The XGBoost based tax model matched reference values well and performed well on outliers, supporting the use of gradient boosting for environmental tabular datasets [20]. Overall, the hybrid AI and deterministic framework delivered realistic billing results while keeping a clear link between penalties or discounts and user behavior.

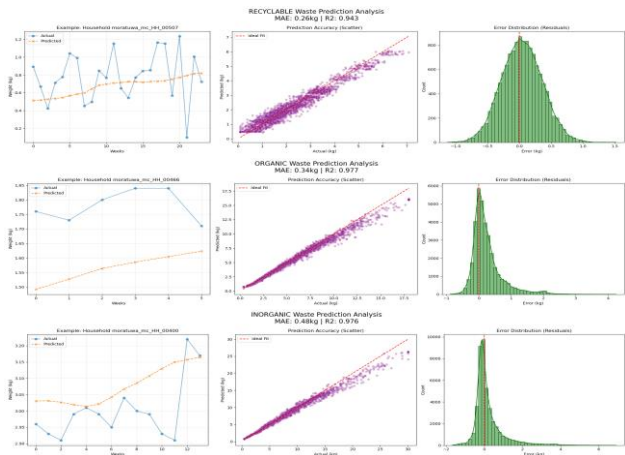


Figure 3 Model Waste Weight Prediction Analysis in three categories

The comparison shown in Figure 3 demonstrates that predicted waste weights closely follow the actual values across the three waste categories used in the smart tax calculation process. Simulations over several months showed that high-waste households consistently faced

higher fees. In contrast, low waste households enjoyed discounts and recognition on the leaderboard. This pattern matches previous gamification research, which suggests that incentives and social comparison can encourage lasting behavior change. [21]. The regression results in Table 2 confirm that the XGBoost tax prediction model produces highly accurate fee estimates, with a strong R^2 value and low prediction errors.

Model	Regression Metrics	Value
Tax prediction (XGBoost)	R^2	0.9989
	RMSE	Rs. 3.68
	MAE	Rs. 0.78
	MAPE	1.85%

Table 2 Tax Prediction Model Metrics Values

For the conversational assistant, cross encoder re-ranking improved the relevance of retrieved document segments, and a hybrid retrieval strategy that combined keyword based and semantic search performed well for Sinhala queries that included location names and collection schedules. Human evaluation further indicated that responses were accurate and aligned with municipal documents when the relevant context was retrieved, while location based filtering increased precision for schedule related queries by restricting candidates to the user's registered area. Overall, grounding responses in official documents reduced the risk of hallucinated rules, and bilingual support improved accessibility. The English RAG evaluation results in Table 3 show that the generated responses have strong lexical and sequence similarity with the reference municipal guidance.

Metric	Mean	Min	Max
BLEU	0.635	0.0	1.0
ROUGEL	0.774	0.087	1.0
BERTScore-F1	0.709	0.075	1.0

Table 3 English RAG evaluation

The English RAG agent achieves higher lexical and sequence overlap (BLEU = 0.635, ROUGE-L = 0.774), while maintaining reasonable semantic similarity (BERTScore-F1 = 0.709), reflecting good performance on English queries with occasional variations in phrasing.

The Sinhala RAG agent demonstrates moderate lexical and sequence overlap with the reference answers (BLEU = 0.513, ROUGE-L = 0.450), while achieving high semantic fidelity (BERTScore-F1 = 0.964), indicating accurate understanding and generation of Sinhala responses. The

Sinhala evaluation in Table 4 shows high semantic similarity, suggesting that the assistant preserves the intended meaning even when the exact word overlap is lower.

Metric	Mean	Min	Max
BLEU	0.513	0.112	1.0
ROUGE-L	0.450	0.0	1.0
BERTScore-F1	0.964	0.910	1.0

Table 4 Sinhala RAG evaluation

For city-scale forecasting, the proposed system applies a temporal LSTM based prediction model to capture historical waste generation patterns within each municipal ward. By learning seasonal trends, short-term demand fluctuations, and environmental influences from historical operational data, the model improves the identification of high risk wards that require immediate attention in municipal collection planning. Event driven case analyses further indicated that the model effectively captures sudden increases in service demand caused by factors such as complaint surges, missed collections, or adverse weather conditions, enabling more reliable detection of operational hotspots. These forecasts are integrated into a municipal planning dashboard that supports exploration of alternative resource allocation and route planning scenarios through an intuitive interface for decision-makers.

The accuracy pattern shown in Figure 4 suggests that the LSTM model learns temporal waste generation patterns that can support ward-level forecasting and municipal resource planning



Figure 4 Model Training Accuracy Graph

The LSTM demand prediction results in Table 5 indicate that the model can classify ward-level waste demand risk with acceptable accuracy, precision, recall, and F1-score values.

Task/Model	Metric	Value
Waste demand prediction (LSTM)	Accuracy	0.781
	Precision (Marco Avg)	0.781
	Recall (Marco Avg)	0.747
	F1-Score (Marco Avg)	0.759
	Best Validation (Marco Avg)	0.830

Table 5 LSTM Waste Demand Prediction Metrics

V. CONCLUSION

This paper presents an integrated smart waste management framework that unifies multimodal artificial intelligence, predictive fee modeling, conversational assistance, and spatiotemporal forecasting on a single platform for sustainable urban waste management. In contrast to many existing approaches that address only isolated functions such as waste classification, forecasting [16], [17], [20], or taxation [19], The proposed framework integrates disposal guidance, behavior-based fee estimation, citizen support, and municipal planning into a coherent, interoperable system.

The framework addresses key limitations in prior work through four tightly connected components. The first component integrates multimodal waste classification [12], disposal recommendation, and IoT-enabled bin monitoring [13], [14], and overflow forecasting to improve the accuracy and timeliness of waste handling operations [15]. The second component introduces a hybrid AI--deterministic fee model that combines LSTM and XGBoost with behavior-aware taxation [16], [17] missing-value handling, and gamification features to promote sustainable user behavior [21]. The third component provides a multilingual retrieval-augmented conversational assistant grounded in official municipal documents, thereby enabling accurate, localized, and user-accessible waste-management guidance [22], [23], [24]. The fourth component supports municipal decision-making through ward-level forecasting, demand scoring, resource allocation, and route optimization.

Taken together, the results indicate that the proposed framework improves waste identification, overflow prediction, transparent fee estimation, citizen engagement, multilingual information access, and operational planning. More broadly, the study demonstrates that AI, IoT, machine learning, and retrieval-augmented conversational systems can be integrated into a scalable and practical smart waste management infrastructure. This integrated

approach offers a promising foundation for more efficient, citizen-centered, and environmentally sustainable waste-management systems in smart cities.

VI. FUTURE WORK

Future work will focus on piloting the system in partner municipalities and collecting longitudinal, anonymized operational data to further calibrate models and evaluate fairness impacts. Additional improvements include expanding localized training data to enable more robust classification, strengthening the conversational assistant through improved document updating and citation, and linking demand forecasting to route optimization and traffic-aware scheduling. Establishing standardized interfaces and governance mechanisms will also be important for responsible integration with other smart-city services.

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